Michael S. Fischer

Assistant Professor of Atmospheric Sciences, University of Miami
Rosenstiel School of Marine, Atmospheric, and Earth Science
Core Faculty, Frost Institute for Data Science and Computing

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RESEARCH Tropical cyclone structure and intensity change · rapid intensification · vortex tilt and alignment · airborne Doppler radar · satellite remote sensing · machine learning

EDUCATION

Ph.D., Atmospheric Science 2018
University at Albany, State University of New York
Dissertation: *Tropical cyclone rapid intensification in environments of upper-tropospheric troughs: Environmental influences and convective characteristics.* Advisors: Brian H. Tang and Kristen L. Corbosiero

B.S., Geosciences (Atmospheric Science), magna cum laude 2013
Florida International University

APPOINTMENTS

Assistant Professor, Department of Atmospheric Sciences 2024–present
Rosenstiel School of Marine, Atmospheric, and Earth Science, University of Miami
Joint appointment: Core Faculty, Frost Institute for Data Science and Computing

Associate Scientist 2023–2024
Cooperative Institute for Marine and Atmospheric Studies (CIMAS), University of Miami

Assistant Scientist 2020–2023
Cooperative Institute for Marine and Atmospheric Studies (CIMAS), University of Miami

National Research Council Postdoctoral Research Associate 2018–2020
NOAA Atlantic Oceanographic and Meteorological Laboratory, Hurricane Research Division

Scientific Programmer 2018
Innovim, at the NOAA National Hurricane Center

HONORS AND AWARDS

Editor's Award, *Advances in Atmospheric Sciences* 2026
Editor's Award, *Journal of the Atmospheric Sciences* 2024
Reviewer's Certificate, *Quarterly Journal of the Royal Meteorological Society* Prize 2020
Distinguished Dissertation Award, University at Albany 2019
Narayan R. Gokhale Distinguished Research Scholarship Award, University at Albany 2018
Outstanding Student Oral Presentation Award, 33rd Conference on Hurricanes and Tropical Meteorology 2018
Top graduate student forecaster for Green Bay, WI, national WxChallenge competition 2015

RESEARCH FUNDING

Current

Application of machine learning techniques to generate three-dimensional tropical cyclone wind analyses, including surface wind estimates
NOAA Weather Program Office, Testbeds (NOAA-OAR-WPO-2025-28601) · **PI**, with Co-PI G. R. Alvey III and Co-Is A. Aksoy and S. D. Ditchek
2026–2029
$747,137

Collaborative Research: Tropical cyclones from 400 to 40 hPa
National Science Foundation · **Co-PI**, with D. S. Nolan and M. E. O'Neill
2024–2027
$777,109

Completed

A comprehensive observational examination of the physical processes that link tropical cyclone vortex alignment to future intensity change
National Science Foundation · **Co-PI**, with G. R. Alvey III and D. S. Nolan
2023–2026
$539,622

Application of machine learning techniques to the automated quality control of airborne Doppler radar data used in operational analysis of hurricane winds and reflectivity
NOAA Weather Program Office (NOAA-OAR-WPO-2023-2007516) · **PI**, with Co-PI M. M. Bell
2023–2025
$400,686

PUBLICATIONS

Refereed journal articles

20. **Fischer, M. S.**, G. R. Alvey III, D. Jariwala, and P. D. Reasor, 2026: To align or not to align? That is the question. *J. Geophys. Res. Atmos.*, **131**, e2025JD045986.
19. Nolan, D. S., **M. S. Fischer**, and M. E. O'Neill, 2025: Reconsideration of the mass and condensate sources for the tropical cyclone outflow. *Bull. Amer. Meteor. Soc.*, **106**, E1342–E1359.
18. **Fischer, M. S.**, P. D. Reasor, J. P. Dunion, and R. F. Rogers, 2025: Are rapidly intensifying tropical cyclones associated with unique vortex and convective characteristics? *Mon. Wea. Rev.*, **153**, 183–203.
17. Rios-Berrios, R., P. M. Finocchio, J. J. Alland, X. Chen, **M. S. Fischer**, S. Stevenson, and D. Tao, 2024: A review of the interactions between tropical cyclones and environmental vertical wind shear. *J. Atmos. Sci.*, **81**, 713–741.
16. Shimada, U., P. D. Reasor, R. F. Rogers, **M. S. Fischer**, F. Marks, J. Zawislak, and J. Zhang, 2024: Shear-relative asymmetric kinematic characteristics of intensifying hurricanes as observed by airborne Doppler radar. *Mon. Wea. Rev.*, **152**, 491–512.
15. **Fischer, M. S.**, R. F. Rogers, P. D. Reasor, and J. P. Dunion, 2024: An observational analysis of the relationship between tropical cyclone vortex tilt, precipitation structure, and intensity change. *Mon. Wea. Rev.*, **152**, 203–225.
14. Wadler, J. B., J. J. Cione, R. F. Rogers, and **M. S. Fischer**, 2023: On the distribution of convective and stratiform precipitation in tropical cyclones from airborne Doppler radar and its relationship to intensity change and environmental wind shear direction. *Mon. Wea. Rev.*, **151**, 3209–3233.
13. Wadler, J. B., J. E. Rudzin, B. J. de la Cruz, J. Chen, **M. S. Fischer**, G. Chen, N. Qin, B. Tang, and Q. Li, 2023: A review of recent research progress on the effect of external influences on tropical cyclone intensity change. *Trop. Cyclone Res. Rev.*, **12**, 200–215.
12. DesRosiers, A. J., M. M. Bell, P. J. Klotzbach, **M. S. Fischer**, and P. D. Reasor, 2023: Observed relationships between tropical cyclone vortex height, intensity, and intensification rate. *Geophys. Res. Lett.*, **50**, e2022GL101877.
11. Stone, Z., G. R. Alvey III, J. P. Dunion, **M. S. Fischer**, D. J. Raymond, R. F. Rogers, S. Sentić, and J. Zawislak, 2023: Thermodynamic contribution to vortex alignment and rapid intensification of Hurricane Sally (2020). *Mon. Wea. Rev.*, **151**, 931–951.

10. **Fischer, M. S.**, P. D. Reasor, B. H. Tang, K. L. Corbosiero, R. D. Torn, and X. Chen, 2023: A tale of two vortex evolutions: Using a high-resolution ensemble to assess the impacts of ventilation on a tropical cyclone rapid intensification event. *Mon. Wea. Rev.*, **151**, 297–320.
9. Hazelton, A., G. J. Alaka, **M. S. Fischer**, R. D. Torn, and S. Gopalakrishnan, 2023: Factors influencing the track of Hurricane Dorian (2019) in the West Atlantic: Analysis of a HAFS ensemble. *Mon. Wea. Rev.*, **151**, 175–192.
8. **Fischer, M. S.**, P. D. Reasor, R. F. Rogers, and J. F. Gamache, 2022: An analysis of tropical cyclone vortex and convective characteristics in relation to storm intensity using a novel airborne Doppler radar database. *Mon. Wea. Rev.*, **150**, 2255–2278.
7. Alvey, G. R., III, **M. S. Fischer**, P. D. Reasor, J. Zawislak, and R. F. Rogers, 2022: Processes underlying the vortex repositioning during Dorian's (2019) early stages that increased its favorability for rapid intensification. *Mon. Wea. Rev.*, **150**, 193–213.
6. Zawislak, J., R. Rogers, S. Aberson, G. Alaka, G. Alvey, A. Aksoy, L. Bucci, J. Cione, N. Dorst, J. Dunion, **M. S. Fischer**, J. Gamache, S. Gopalakrishnan, A. Hazelton, H. Holbach, J. Kaplan, H. Leighton, F. Marks, S. Murillo, P. Reasor, K. Ryan, K. Sellwood, J. Sippel, and J. Zhang, 2022: Accomplishments of NOAA's Airborne Hurricane Field Program and a broader future approach to forecast improvement. *Bull. Amer. Meteor. Soc.*, **103**, E311–E338.
5. Hazelton, A., G. J. Alaka, L. Cowan, **M. S. Fischer**, and S. Gopalakrishnan, 2021: Understanding the processes causing the early intensification of Hurricane Dorian through an ensemble of the Hurricane Analysis and Forecast System (HAFS). *Atmosphere*, **12**, 93.
4. **Fischer, M. S.**, R. F. Rogers, and P. D. Reasor, 2020: The rapid intensification and eyewall replacement cycles of Hurricane Irma (2017). *Mon. Wea. Rev.*, **148**, 981–1004.
3. **Fischer, M. S.**, B. H. Tang, and K. L. Corbosiero, 2019: A climatological analysis of tropical cyclone rapid intensification in environments of upper-tropospheric troughs. *Mon. Wea. Rev.*, **147**, 3693–3719.
2. **Fischer, M. S.**, B. H. Tang, K. L. Corbosiero, and C. M. Rozoff, 2018: Normalized convective characteristics of tropical cyclone rapid intensification events in the North Atlantic and eastern North Pacific. *Mon. Wea. Rev.*, **146**, 1133–1155.
1. **Fischer, M. S.**, B. H. Tang, and K. L. Corbosiero, 2017: Assessing the influence of upper-tropospheric troughs on tropical cyclone intensification rates after genesis. *Mon. Wea. Rev.*, **145**, 1295–1313.

Submitted

Fischer, M. S., E. Danto, D. S. Nolan, W. G. Bouza, and E. Siman: Mid-level outflow in tropical cyclones observed by airborne Doppler radar. *Mon. Wea. Rev.*, submitted.

In preparation

Fischer, M. S., W. G. Bouza, D. Jariwala, and M. Reardon: Estimating tropical cyclone vortex tilt from geostationary satellite imagery using a random forest. *Wea. Forecasting*.

Alvey, G. R., III, J. Zhang, J. Dunion, B. Dahl, and **M. S. Fischer**: The path to alignment: Dual-Doppler and SAMURAI observations of Tropical Cyclone Fiona (2022). *Mon. Wea. Rev.*

Book chapters

Sippel, J. A., and **M. S. Fischer**, 2026: Tropical cyclone intensity predictability. *Encyclopedia of Atmospheric Sciences*, 3rd ed., Vol. 1, Elsevier, 446–459.

RESEARCH DATA AND SOFTWARE

TC-RADAR, Tropical Cyclone Radar Archive of Doppler Analyses with Recentering. Developer and data steward of a database of more than 1,600 airborne Doppler radar analyses from storms sampled by NOAA's WP-3D aircraft, 1997–2025. 2019–present

TC-ATLAS, tcatlas.org. Developer of an open-source, interactive web platform for real-time monitoring and historical analysis of tropical cyclone structure and intensity change. 2025–present

PRESENTATIONS

Invited talks and seminars

Fischer, M. S.: Seeing the unseen: Revealing tropical cyclone structure with observations and AI. DAES/ASRC Joint Colloquium, University at Albany, Albany, NY, November 2026. INVITED

Fischer, M. S.: Tropical cyclone structure and intensity change: What observations reveal, and how machine learning fills the gaps. COMPASS Seminar, October 2026.

Fischer, M. S.: A machine learning approach to estimate the three-dimensional tropical cyclone kinematic structure. NSF IPAM Workshop on Mathematics and Machine Learning for Earth System Simulation, Los Angeles, CA, February 2026. INVITED

Fischer, M. S.: A machine learning approach to estimate the three-dimensional tropical cyclone wind field. AGU Annual Meeting, New Orleans, LA, December 2025. INVITED

Fischer, M. S.: Exploring tropical cyclone structural precursors to rapid intensification. Symposium for Tropical Cyclone Risk in a Changing Climate, Clearwater, FL, June 2025. INVITED

Fischer, M. S.: Using machine learning to better understand tropical cyclone structure and intensity change. Frost Institute for Data Science and Computing Seminar, University of Miami, Miami, FL, April 2025.

Fischer, M. S.: Data-driven insights into tropical cyclone rapid intensification. Department seminar, University of Alabama in Huntsville, Huntsville, AL, March 2025.

Fischer, M. S., P. D. Reasor, R. F. Rogers, J. P. Dunion, B. H. Tang, K. L. Corbosiero, R. D. Torn, and X. Chen: Exploring the relationship between vortex misalignment and tropical cyclone intensity change. Hurricane Forecast Improvement Project Seminar, April 2023.

Conference presentations

Fischer, M. S., W. G. Bouza, D. Jariwala, and M. Reardon: A machine learning approach to estimate the three-dimensional tropical cyclone wind field. 37th Conf. on Hurricanes and Tropical Meteorology, San Diego, CA, April 2026.

Fischer, M. S.: A machine learning approach to estimate tropical cyclone structure using geostationary satellite observations. Working Group on Tropical Meteorology Research Annual Meeting, virtual, May 2025.

Fischer, M. S., P. D. Reasor, J. P. Dunion, and R. F. Rogers: Are rapidly intensifying tropical cyclones associated with unique vortex characteristics? 36th Conf. on Hurricanes and Tropical Meteorology, Long Beach, CA, May 2024.

Fischer, M. S., R. F. Rogers, P. D. Reasor, and J. P. Dunion: The relationship between tropical cyclone vortex tilt, precipitation structure, and intensity change. 104th AMS Annual Meeting, Baltimore, MD, January 2024.

Fischer, M. S., P. D. Reasor, R. F. Rogers, and J. F. Gamache: An analysis of tropical cyclone vortex and convective characteristics in relation to storm intensity using a novel airborne Doppler radar database. 35th Conf. on Hurricanes and Tropical Meteorology, New Orleans, LA, May 2022.

Fischer, M. S., R. F. Rogers, P. D. Reasor, and J. P. Dunion: Relationships between vortex tilt, convective structure, and intensity change in early-stage tropical cyclones. 34th Conf. on Hurricanes and Tropical Meteorology, virtual, May 2021.

Fischer, M. S., R. F. Rogers, P. D. Reasor, and J. P. Dunion: How is tropical cyclone vortex tilt related to precipitation structure and intensity change? 14th International Conf. on Mesoscale Convective Systems and High-Impact Weather in East Asia, virtual, April 2021.

Fischer, M. S., R. F. Rogers, and P. D. Reasor: An examination of local shear, vortex tilt, and tropical cyclone intensity change using airborne radar observations. 100th AMS Annual Meeting, Boston, MA, January 2020.

Fischer, M. S., R. F. Rogers, and P. D. Reasor: The rapid intensification and eyewall replacement cycles of Hurricane Irma (2017). 19th Cyclone Workshop, Seeon, Germany, October 2019.

Fischer, M. S., B. H. Tang, and K. L. Corbosiero: Characteristics of tropical cyclone rapid intensification in environments of upper-tropospheric troughs. 33rd Conf. on Hurricanes and Tropical Meteorology, Ponte Vedra Beach, FL, April 2018.

Fischer, M. S., B. H. Tang, and K. L. Corbosiero: Convective characteristics of tropical cyclone rapid intensification in environments of upper-tropospheric troughs. 8th Northeast Tropical Meteorology Workshop, Rensselaerville, NY, June 2017.

Fischer, M. S., B. H. Tang, and K. L. Corbosiero: The influence of an upper-tropospheric potential vorticity anomaly on rapid tropical cyclogenesis. 32nd Conf. on Hurricanes and Tropical Meteorology, San Juan, Puerto Rico, April 2016.

TEACHING

University of Miami, instructor

RSM 780: Machine Learning Applications in the Atmospheric Sciences (graduate)	Fall 2026
ATM 102: Introduction to Weather and Climate	Fall 2025
RSM 780: Directed Readings in Tropical Weather and Climate (graduate)	Spring 2025
ATM 411/412: Research in Atmospheric Science (undergraduate thesis)	2025–2026

Guest lectures

ATM 652: Introduction to Atmospheric Physics, University of Miami	2024, 2025
ATM 536: Hurricanes, University of Miami	2024
ATM 303: Meteorological Instrumentation and Observation, University of Miami	2022, 2025
Weather Discussion I, St. Cloud State University	

Teaching assistant, University at Albany

Dynamic Meteorology I and II; Tropical Meteorology; Natural Disasters; Understanding the Earth	2013–2016

ADVISING AND MENTORING

Graduate students, primary advisor

Kelly Neighbour, Ph.D., University of Miami	2025–present
Helena Tsigos, Ph.D., University of Miami	2025–present
Cameron Pine, Ph.D., University of Miami. Co-advised with D. S. Nolan and G. R. Alvey III	2023–present
Shannon McCloskey, M.P.S., University of Miami. Thesis: *An observational analysis of rapidly intensifying tropical cyclones undergoing eyewall replacement cycles*	2026

Amanda Keane, M.P.S., University of Miami. Thesis: *Using NOAA Tail Doppler Radar observations to examine the evolution of vortex tilt in rapidly intensifying Hurricane Ida (2021)* — 2022

Graduate committees

Aidan Mahoney, Ph.D., University of Miami — Defended 2026
Matthew Donahue, M.S., University of Miami — Defended 2026

Undergraduate thesis advisor

Arielle Jorge, University of Miami — 2025–present
Deelan Jariwala, University of Miami — 2026
Matthew Reardon, University of Miami — 2026

SERVICE

University

Chair, Rosenstiel Award Committee (2027 award), Rosenstiel School — 2026–2027

Editorial and review

Associate Editor, *Monthly Weather Review* — 2020–present
Proposal reviewer, National Science Foundation and NASA — 2023–present
Journal reviewer: *J. Atmos. Sci.*, *Mon. Wea. Rev.*, *J. Appl. Meteor. Climatol.*, *Quart. J. Roy. Meteor. Soc.*, *J. Geophys. Res. Atmos.*, *Geophys. Res. Lett.*, *npj Climate Atmos. Sci.*, *Wea. Climate Dyn.*, *Dyn. Atmos. Oceans*, *Atmosphere* — 2017–present

Research programs

NOAA Hurricane Research Division Field Program science team; PI of the Vortex Alignment Module — 2018–present
Rapid Intensification in Tropical Cyclones (TCRI) science team — 2020–2024
NOAA Hurricane and Ocean Testbed collaborator — 2021–2024

Conferences and workshops

Rapporteur, 11th WMO International Workshop on Tropical Cyclones — 2026
Session co-convener, 37th Conf. on Hurricanes and Tropical Meteorology — 2025–2026
Session chair, Tropical Cyclone Rapid Intensification, 35th–37th Conf. on Hurricanes and Tropical Meteorology — 2022–2026
Working group member, 10th WMO International Workshop on Tropical Cyclones — 2022
Max Eaton Award committee, 34th and 35th Conf. on Hurricanes and Tropical Meteorology — 2021, 2022
Student poster award committee, 34th Conf. on Hurricanes and Tropical Meteorology — 2022
Session co-chair, Rapid Intensification of Tropical Cyclones, 101st AMS Annual Meeting — 2021

Outreach

“A career in hurricane science,” South Miami Heights Elementary School — 2024–present

PROFESSIONAL MEMBERSHIPS

American Meteorological Society (2014–present) · American Geophysical Union (2025–present)